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Arduino UNO R4 WiFi ADC Resolution

Learn how to change the ADC resolution on the UNO R4 WiFi.

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In this tutorial you will learn how to change the analog-to-digital converter (ADC) on an **Arduino UNO R4 WiFi** board. By default, the resolution is set to 10-bit, which can be updated to both 12-bit (0-4096) and 14-bit (0-16383) resolutions for improved accuracy on analog readings.

Goals

The goals of this tutorials are:

Update the ADC resolution to 12/14-bit.

Hardware & Software Needed

Arduino IDE ([online](#) or [offline](#))

[Arduino UNO R4 WiFi UNO R4 Board Package](#)

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Analog-to-Digital Converter (ADC)

An analog-to-digital converter (ADC) transforms an analog signal to a digital one. The standard resolution on Arduino boards is set to 10-bit (0-1023). The UNO R4 WiFi supports up to 14-bit resolutions, which can provide a more precise value from analog signals.

To update the resolution, you will only need to use the `analogReadResolution()` command.

To use it, simply include it in your `setup()`, and use `analogRead()` to retrieve a value from an analog pin.

```
1 void setup(){
2   analogReadResolution(
3 }
4
5 void loop(){
6   int reading = analogR
7 }
```

Summary

This short tutorial demonstrates how to update the resolution for your ADC, a new feature available on the UNO R4 WiFi board.

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